Explainable Deep Learning for Bone Marrow Cell Classification Using EfficientNet-B0 and Grad-CAM

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***Abstract*—The diagnosis of hematologic malignancies requires bone marrow cytology; though, manual smear inspection is still unreliable, time-consuming, and often error-prone. In order to automatically categorize five clinically relevant bone marrow cell types, promyelocyte (PMO), myeloblast (BLA), neutrophilic granulocyte (NGB), plasma cell (PLM), and myelocyte (MYB), this study suggests a Deep Learning (DL) based approach. Training four Convolution Neural Network (CNN) architectures, EfficientNet-B0, DenseNet121, MobileNetV2, and ResNet50, involves preprocessing and augmenting a carefully selected Kaggle dataset. With 90.28% accuracy, a macro F1-score of 0.899, and ROC AUC values over 0.99 for the majority of classes, EfficientNet-B0 outperformed the others. By emphasizing clinically significant nuclear and cytoplasmic areas, Grad-CAM visualizations improved interpretability and clinical confidence. These findings show that computational techniques can benefit hematopathologists by simplifying diagnostic tasks, enhancing consistency, and facilitating quicker, more accurate decision-making in daily practice.**

***Index Terms*—Bone marrow cytology, deep learning, convolutional neural networks, medical image classification, EfficientNet, Grad-CAM**

I. Introduction

Bone marrow is a soft, spongy substance that is located in the cavities of bones, producing the blood cells necessary to maintain human life [1]. It produces platelets to aid in blood vessel coagulation, white blood cells to aid in immunologic defense, and red blood cells to carry oxygen. The immune system's stability and the early detection of pathological alterations depend heavily on white blood cell precursors, including PMO, BLA, NGB, PLM, and MYB [2]. Multiple myeloma, leukemia, lymphoma, and myeloproliferative neoplasms are among the hematological illnesses that are closely linked to abnormalities in the morphology, proportion, or distribution of these cells. Therefore, in clinical practice, rapid diagnosis, treatment planning, and prognosis depend on the accurate categorization of bone marrow cells [2], [3].

Despite its significance, hematologists have historically identified bone marrow cells by examining stained bone marrow smears under a microscope [4]. Despite being the gold standard, this manual inspection involves subjectivity and inter-observer variability because it heavily relies on the observer's competence. Furthermore, the diagnostic procedure is sometimes lengthy and involves substantial training, especially when working with uncommon or morphologically related cell types. These difficulties might result in mistakes and delays in diagnosis in high-burden clinical settings, which will eventually impact patient care [5], [6].

To develop methods to help hematologists reduce effort and improve diagnostic consistency, experts have recently investigated a variety of computer-aided diagnostic (CAD) systems and Machine Learning (ML) techniques [7]. Medical image analysis, including radiography, cytology, and histopathology, has benefited greatly from recent developments in Artificial Intelligence (AI), especially DL [8]. In order to improve diagnosis accuracy and eliminate the need for human experience, CNNs have shown exceptional capacity to extract discriminative characteristics from complicated biological pictures automatically. DL approaches can greatly improve the accuracy and efficiency of cell categorization in bone marrow cytology, which will ultimately help hematologists make quicker and more reliable diagnoses.

The main contribution of this paper is:

- We presented a strong DL system that uses a sizable, enhanced dataset to categorize five different kinds of bone marrow cells.
- After comparing four D-CNNs designs, EfficientNet-B0 was found to be the most successful model, with good generalization of 90.28% accuracy.
- Grad-CAM visualizations improve interpretability by connecting model predictions to morphological characteristics that are clinically significant.

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The remainder of this paper is structured as follows: section II presents a broad review of existing related work in bone marrow cell; in section III, we describe the methodology proposed: data preprocessing, model architecture, and optimization techniques. In section IV, we present and discuss our experimental results, benchmarking against other state-of-the-art methods. Finally, section V concludes this paper, with future research directions outlined.

II. Literature Review

The diagnosis of hematologic malignancies relies heavily on bone marrow cytology; still, manual smear examination is difficult, subjective, and prone to inconsistencies. Interest in AI, particularly DL, to improve accuracy and automate categorization has increased as a result of these constraints. There are still issues, nevertheless, such as model interpretability, inconsistent picture quality, imbalanced datasets, and a lack of publicly available datasets. Specialized neural architectures, sophisticated preprocessing, synthetic data augmentation, and multimodal techniques are some of the solutions that researchers have suggested to solve these issues.

Tarquino et al. [1] proposed a region-attention embedding approach that uses Xception and ResNet50 backbones to arrange cytological information into spatial matrices, with an F1-score of 0.82 across 21 BM subtypes and interpretability. Chen et al. [2] expanded on this by introducing SCKansformer, which combines global-local encoding, spatial-channel reconstruction, and kernel attention. Their method outperformed traditional CNNs and set a standard for subsequent studies after being evaluated on a fresh dataset with around 40 classifications. Wang et al. [3] made considerable progress with DCCBANet, a dual-attention network that uses Dynamic Focal Loss and dilated convolutions. The network reported macro F1-scores of 87.49% for granulocytes and 96.8% for blasts.

Likewise, Atasoy et al. [4] experimented with upgraded Capsule Networks (CapsNets), which incorporate dynamic routing. Their GN-CapsNet achieved up to 99.24% accuracy on a 21-class dataset by rebalancing classes using SMOTE. Wang et al. [9] used CW-Net, a weakly supervised model that achieved nearly flawless accuracy and recall, to address annotation issues in whole-slide image analysis. In the meantime, Tarmissi et al. [5] created MTLSO-BMCC, a multimodal hybrid framework that combines an optimized extreme learning machine with three CNNs, resulting in fast inference and 98.6% accuracy. Eckardt et al. [6] used StyleGAN2-Ada to create artificial AML and APL pictures in order to overcome the lack of data. When added to training, their outputs enhanced model performance and were convincing enough to pass a Turing test given by a hematologist.

Furthermore, after conducting a thorough study of DL techniques for BM cytology, Ghete et al. [10] identified many major challenges, including class imbalance, varied datasets, and poor repeatability. The necessity of interpretability criteria and macro-averaged metrics was underlined.

III. Methodology

For accurate bone marrow cell type disease detection, the methodology includes data collection, preprocessing, model training, and evaluation. The entire process is shown in Figure 1.

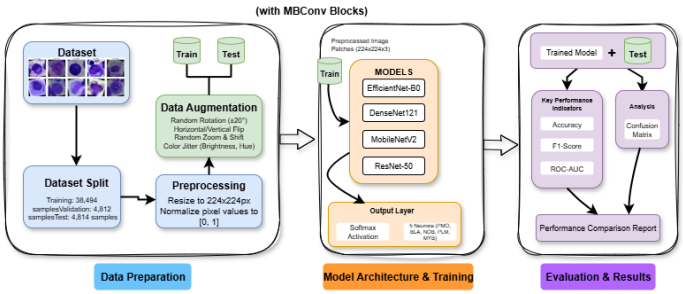
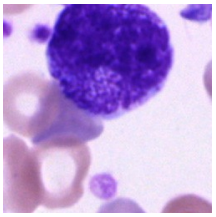


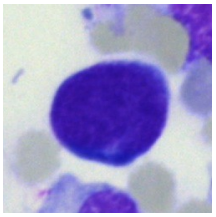
Fig. 1: Overall workflow of our research

A. Dataset Description

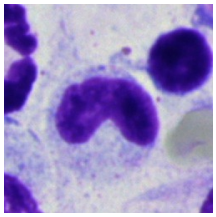
The study's dataset, Figure 2, which was obtained via Kaggle [11], includes microscopic pictures of five clinically significant bone marrow cell types: myeloblast (BLA), promyelocyte (PMO), neutrophil granulocyte (NGB), plasma cell (PLM), and myelocyte (MYB). In order to identify hematologic malignancies like leukemia and multiple myeloma, these cells must be accurately identified since they represent crucial phases of hematological development.



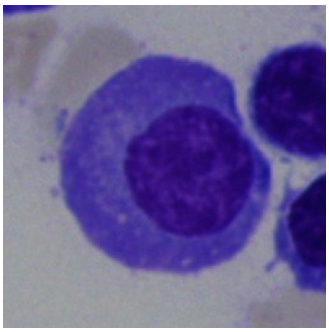
(a) Promyelocyte



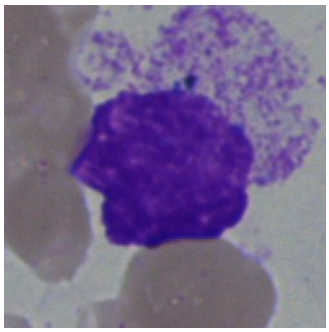
(b) Myeloblast



(c) Neutrophilic Granulocyte



(d) Plasma Cell



(e) Myelocyte

Fig. 2: Samples from the dataset

Each class has unique morphological characteristics that frequently overlap. As an instance, neutrophilic granulocytes have segmented nuclei and granular cytoplasm, whereas myeloblasts have large nuclei with visible nucleoli. Though

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myelocytes and promyelocytes seem as intermediate precursors with minor variations, plasma cells are distinguished by their eccentric nuclei and basophilic cytoplasm.

B. Data Preprocessing & Augmentation

To improve data quality and model generalization, here we preprocessed our datasets and augmented them. The Original datasets are 20,186 microscopic pictures of bone marrow cells were preprocessed. In order to stabilize training, preprocessing operations included scaling images to a consistent resolution appropriate for deep learning models, normalizing pixel intensities, and, if required, removing noise or artifacts. Extensive augmentation techniques were used to resolve potential class imbalance and boost data variety, resulting in a final collection of 42,186 images. These enhancements included intensity-based changes like brightness and contrast modification, as well as geometric changes like rotation, flipping, scaling, and shifting in both the horizontal and vertical directions.

TABLE I: Total Number of Train-Test Images From Original and Augmented Images

Dataset Type	Total Images	Training Images	Validation Images
Preliminary Dataset	25,000	20,186	4,814
Augmented Dataset	47,000	42,186	4,814

Keeping the train and test sets ratio at 70:30, the models were able to learn more robust and invariant characteristics from bone marrow smear visualizations through this preprocessing and augmentation, which also increased the dataset and decreased the chance of overfitting.

C. Learning Phases

1) ***EfficientNet-B0****:* EfficientNet-B0 is a CNN that uses compound scaling to balance input resolution, model depth, and width. Compared to conventional CNNs, it uses fewer parameters to obtain remarkable precision [12].

$$e = \alpha^{\phi}, \quad f = \beta^{\phi}, \quad q = \gamma^{\phi} \quad \text{subject to } \alpha \cdot \beta^2 \cdot \gamma^2 \approx 2 \tag{1}$$

Here, e (depth), f (width), and q (resolution) are scaled using constants α, β, γ. The coefficient ϕ controls scaling, ensuring efficient accuracy–resource trade-offs.

2) ***DenseNet-121****:* The CNN DenseNet-121 has feed-forward connections between each layer and all other layers. This connection minimizes vanishing gradients, decreases the number of parameters, and enhances feature reuse [13].

$$y_\ell = F_\ell([x_0, x_1, \ldots, y_{\ell-1}]) \tag{2}$$

Here, y_ℓ is the output of the ℓ-th layer, and $[x_0, x_1, \ldots, y_{\ell-1}]$ denotes concatenation of feature maps from all previous layers. F_ℓ is a composite function of BatchNorm, ReLU, and convolution.

3) ***MobileNetV2****:* A lightweight CNN designed for embedded and mobile devices is called MobileNetV2. It reduces computation while preserving accuracy by using depthwise separable convolutions with inverted residual blocks [14].

$$x = M_{pm}^{(2)} \cdot \sigma\big(M_{dm} \cdot \sigma(M_{pm}^{(1)} \cdot y)\big) \tag{3}$$

Here, $M_{pm}^{(1)}$ and $M_{pm}^{(2)}$ are pointwise (1×1) convolutions, and M_{dm} is a depthwise convolution. The non-linearity σ (ReLU6) allows efficient feature extraction with fewer parameters.

4) ***ResNet50****:* In order to compensate for vanishing gradients, ResNet50, a 50-layer CNN, adds residual connections, also known as skip connections. It makes it possible to train extremely deep networks without sacrificing efficiency [15].

$$y_\ell = F(x_\ell, W_\ell) + x_\ell \tag{4}$$

Here, x_ℓ is the input to the residual block, $F(x_\ell, W_\ell)$ represents the residual mapping (convolution + batch normalization + activation), and the skip connection adds the original input to the output, enabling smooth gradient flow.

IV. Result and Discussion

A. Evaluation Metrics

The performance of the proposed DL models was evaluated using several well-known assessment metrics. Accuracy provided an overall indication of correct predictions, while precision and recall highlighted the model's ability to identify bone marrow cells and reduce false positives. The confusion matrix, presenting true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), offered a more detailed view of classification performance. Finally, the F1 Score was used to measure the balance between precision and recall.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{8}$$

B. Performance Evaluation

As shown in Table II, the performance of the suggested models was thoroughly examined using important classification metrics, including precision, recall, F1-score, and total accuracy. EfficientNet-B0 outperformed the other applicable designs in classifying between bone marrow cell types, with the greatest overall accuracy of 90.28%. Its ability to persevere in capturing discriminative behaviors is demonstrated by the consistently high results it produced throughout the majority of categories, especially for NGB Precision: 96%, Recall: 97%, F1-score: 96 and PLM Precision: 93%, Recall: 98%, F1-score: 95.

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TABLE II: Performance results for all applied models

Model	Class	Precision (%)	Recall (%)	F1-score (%)	Accuracy (%)
EfficientNet-B0	PMO	94	92	93	**90.28**
	BLA	73	87	80	
	NGB	96	97	96	
	PLM	93	98	95	
	MYB	91	80	85	
DenseNet121	PMO	94	93	89	**89.43**
	BLA	80	81	83	
	NGB	92	97	96	
	PLM	92	97	95	
	MYB	91	81	83	
MobileNetV2	PMO	93	93	89	**89.34**
	BLA	80	81	83	
	NGB	93	96	96	
	PLM	92	97	95	
	MYB	91	81	83	
ResNet50	PMO	93	92	91	**88.39**
	BLA	73	82	79	
	NGB	92	95	97	
	PLM	91	96	94	
	MYB	88	79	83	

With an accuracy of 89.43%, DenseNet-121 came in next, attaining a balance between precision and recall, particularly for the NGB and PLM classes, where it achieved F1-scores of 96 and 95, respectively. Similarly, MobileNetV2 demonstrated competitive results and high generalization ability while having a lighter architecture, with an accuracy of 89.34%. Even though ResNet50 performed a little better accuracy of 88.39%, it nonetheless generated trustworthy classification results, especially for NGB precision: 92%, recall: 95%, F1-score: 97.

EfficientNet-B0 demonstrated its capacity to learn robust and invariant features from bone marrow smear pictures by outperforming the other models overall in terms of classification accuracy and F1-score. All models, however, continued to perform competitively, confirming deep learning's efficacy in automated bone marrow cell categorization.

C. Error Analysis

The confusion matrix of EfficientNet-B0 is shown in Figure 3 to further analyze model performance. With outstanding results for PMO 1105 accurate, NGB 964 right, PLM 745 correct, and MYB 960 correct, the model correctly categorized most samples in all categories. These findings attest to the model's exceptional ability to identify these cell types' predominant morphological patterns.

Nonetheless, several misclassifications were noted, particularly amongst groups that shared morphology. Because early myeloblast and myelocyte stages share visual characteristics, a minority of MYB cells 170 occurrences were mistakenly categorized as BLA. The infrequent misclassification of PMO samples 43 cases as MYB also suggests feature similarities between nuclear and cytoplasmic elements. These mistakes show the natural challenges of discriminating visually overlapping bone marrow cell groups, even if the misclassification rates were quite low when compared to the properly predicted occurrences.

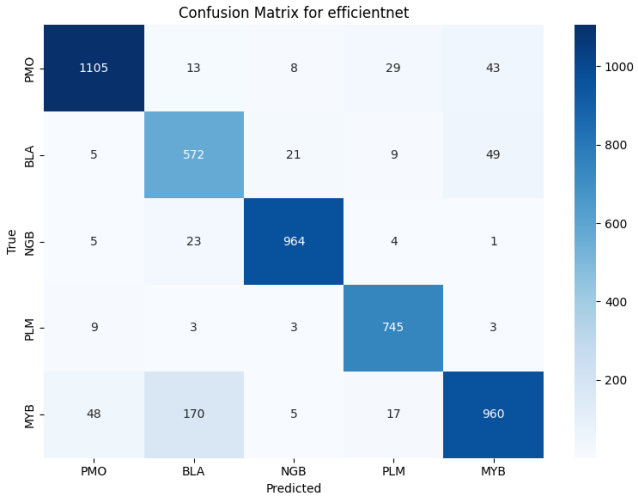


Fig. 3: Illustration of EfficientNet-B0 Confusion matrix for DL models

According to the confusion matrix study, EfficientNet-B0 only has trouble with borderline situations when inter-class similarities are noticeable, maintaining good reliability across all classes overall.

D. ROC Curve & AUC Curve Analysis

All bone marrow cell types show consistently great performance in the class-wise study of EfficientNet-B0, which is shown in Table III and Figure 4. Strong discriminative performance for PLM 0.9961 and NGB 0.9953 was demonstrated by the model's greatest ROC AUC and outstanding average accuracy values of 0.9904 and 0.9899, respectively. With ROC AUC values of 0.9928 and 0.9774, respectively, and average precision of 0.9804 and 0.9369, PMO and MYB also achieved strong scores that demonstrated dependable classification results. The BLA class, on the other hand, performed quite poorly, with an average accuracy of 0.8770, underscoring its greater vulnerability to misclassification.

TABLE III: Class-wise ROC AUC and Average Precision EfficientNetB0

Class	ROC AUC	Avg. Precision
PMO	0.9928	0.9804
BLA	0.9752	0.8770
NGB	0.9953	0.9899
PLM	0.9961	0.9904
MYB	0.9774	0.9369

Overall, EfficientNet-B0 demonstrated a great capacity for generalization across all classes, with NGB and PLM exhibiting a particularly high precision-recall balance. However, small shortcomings in BLA classification indicate that further work is needed to differentiate between morphologically similar cell types.

The ROC curves of all assessed models are shown in Figure 5, emphasizing their capacity to differentiate between different classes of bone marrow cells. With a micro-average

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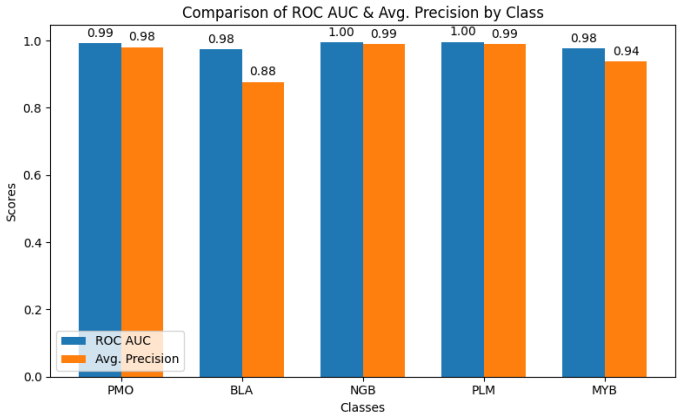
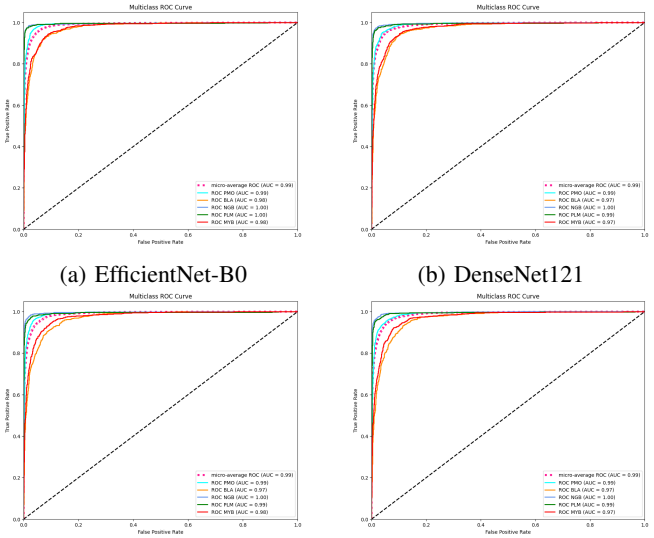


Fig. 4: Illustrate comparison of ROC AUC & Avg. Precision by every class of EfficientNet-B0

AUC of 0.99, EfficientNet-B0 demonstrated the strongest overall separability and superior classification consistency across all classes. With micro-average AUC values of 0.99 each, DenseNet121 and MobileNetV2 demonstrated competitive performance as well, demonstrating their dependability in multi-class classification. Across all models, NGB and PLM consistently obtained near-perfect AUC values of 1.00 among the individual classes, indicating their unique morphological characteristics that are simpler to capture. The BLA and MYB classes, on the other hand, displayed somewhat lower AUC values, especially for ResNet50, suggesting difficulties in differentiating between morphologically similar subtypes.



(a) EfficientNet-B0 (b) DenseNet121

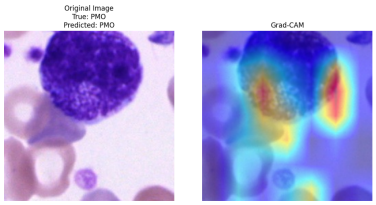
(c) MobileNetV2 (d) ResNet50

Fig. 5: ROC-Curve all models

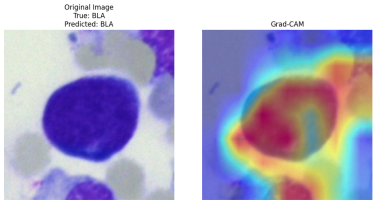
EfficientNet-B0 showed the most balanced and robust discriminative power, demonstrating its promise as the most clinically reliable model for bone marrow cell classification, even though all four models maintained good ROC performance overall.

E. Grad-CAM Visualizations

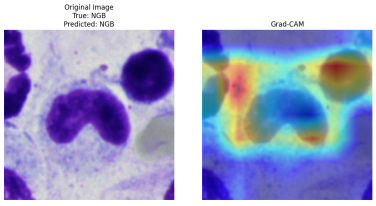
A visualization method known as Gradient-weighted Class Activation Mapping (Grad-CAM) gives CNN-based models interpretability. It uses gradients from the last convolutional layer to highlight areas of the picture that have a significant impact on the model's choice [16].



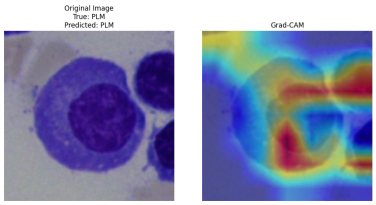
(a) Original PMO and output of Grad_CAM



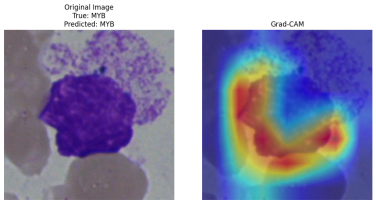
(b) Original BLA and output of Grad_CAM



(c) Original NGB and output of Grad_CAM



(d) Original PLM and output of Grad_CAM



(e) Original MYB and output of Grad_CAM

Fig. 6: Five classes original image and their Grad-CAM visualizations

$$\alpha_k^c = \frac{1}{Z} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k} \tag{9}$$

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$$G^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right) \tag{10}$$

Here, y^c is the score for class c, and A^k is the k-th feature map of the last convolutional layer. The importance weight α^c_k is computed by averaging gradients over spatial dimensions, showing how strongly feature map k contributes to class c. The final heatmap $G^c_{\text{Grad-CAM}}$ is the result of a weighted sum of feature maps, where ReLU retains just the positive outcomes [17].

The Grad-CAM visuals for the five typical classes PMO, BLA, NGB, PLM, and MYB are shown in Figure 6, emphasizing the important picture areas that influenced the model's predictions.

The heatmaps for PMO and NGB cells are primarily focused on the cytoplasmic textures and nuclear borders, suggesting that the model well captures morphological characteristics that are essential for categorization. Although there are infrequent overlaps that indicate more classification difficulties, the highlighted regions in the cases of BLA and MYB cover larger nuclear areas, demonstrating the model's emphasis on tiny differences in nuclear shape and density that distinguish these classes. The display for PLM highlights both cytoplasmic and nuclear areas, demonstrating that the model takes on multi-region relationships, enabling precise identification.

Overall, by showing that the model not only attains high accuracy but also makes its decisions on clinically significant morphological areas, the Grad-CAM results improve interpretability and confidence in the categorization framework.

V. Conclusion

This work demonstrates that deep learning may be used for bone marrow cytology, since EfficientNet-B0 outperforms other CNNs in terms of accuracy, F1-scores, and AUC values. Because Grad-CAM images matched predictions with significant morphological areas, interpretability was significantly enhanced. All of the results together demonstrate that automated categorization can lessen the workload of pathologists, enhance the consistency of diagnoses, and provide more trustworthy hematologic evaluations.

Future studies should use adaptive loss functions, synthetic data creation, and improved augmentation to overcome class imbalance. Incorporating multimodal data (cytology, genomic, and clinical) can increase diagnosis accuracy, while expanding datasets with multi-institutional samples will improve model generalization. Additionally, implementing complex explainable AI techniques beyond Grad-CAM will improve clinician confidence, and real-time implementation in pathology laboratories would allow for useful diagnostic guidance.

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